Project Seminar 2026





Design of a fuel cell monitoring board in KiCad and development of the operational board software

Lesson 10: Development Operational Board Software

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FIW: Fachbereich Elektrotechnik und Informatik

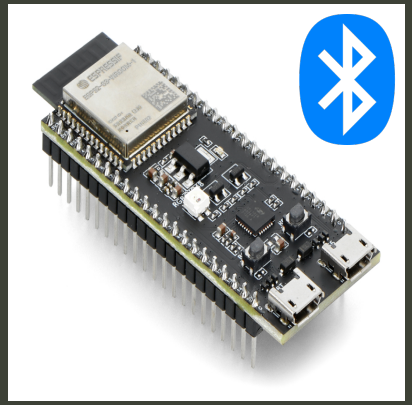
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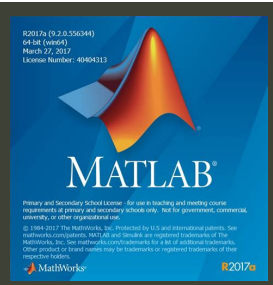
- ***Repetition: Fuel cell monitoring and measurement board setup***







GNU Octave



Fuel cell (500W)

Measurement board with ESP32-S3 micro-controller and DAQ-board software

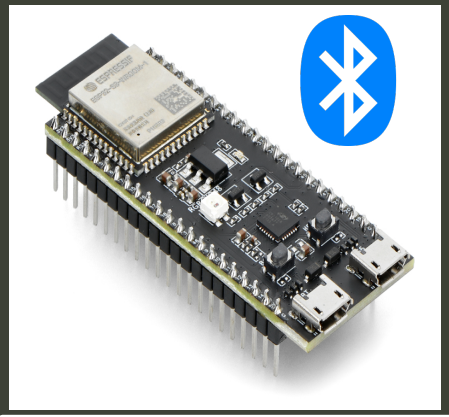
DAQ - Software

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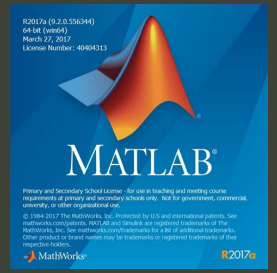
- ***Repetition: System measurement chain (1)***







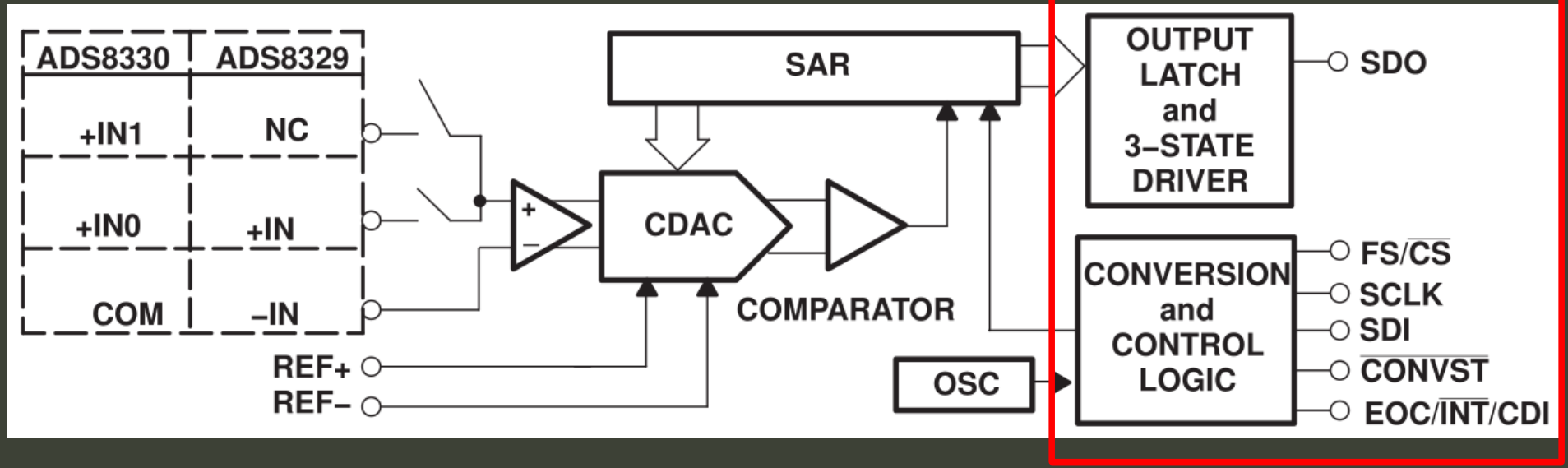
GNU Octave



Physical quantities to be measured	→	Signal preprocessing and signal conditioning	→	Micro-Controller	→	DAQ - Software
→ Output current, → Output voltage		→ Sensing → Amplification → Digitization		→ Data Acquisition → Data Processing → Data Transmission		

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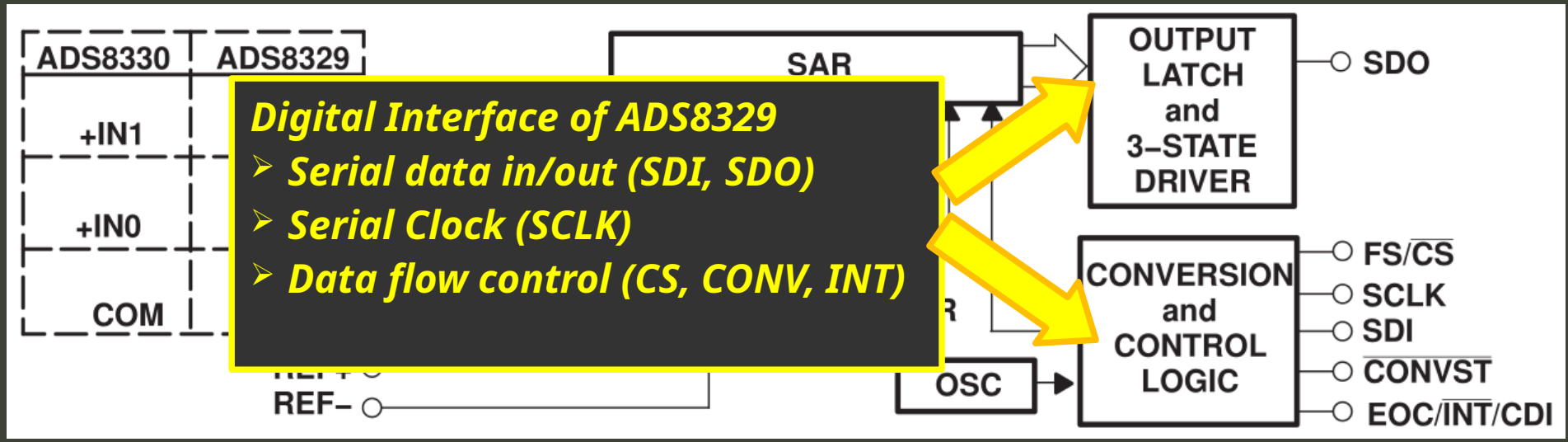
- ***Repetition: ADS8329 → ADC → Communication Interface***



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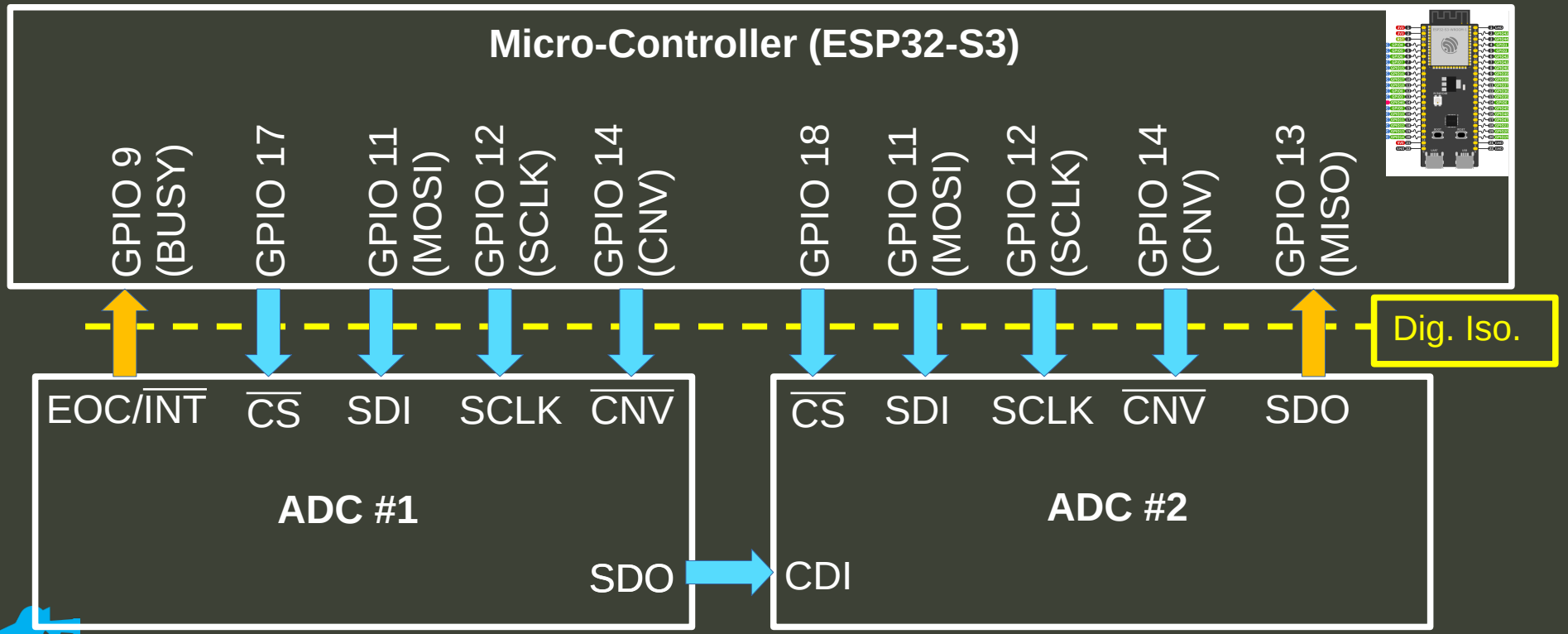


- ***Repetition: ADS8329 → ADC → Communication Interface***



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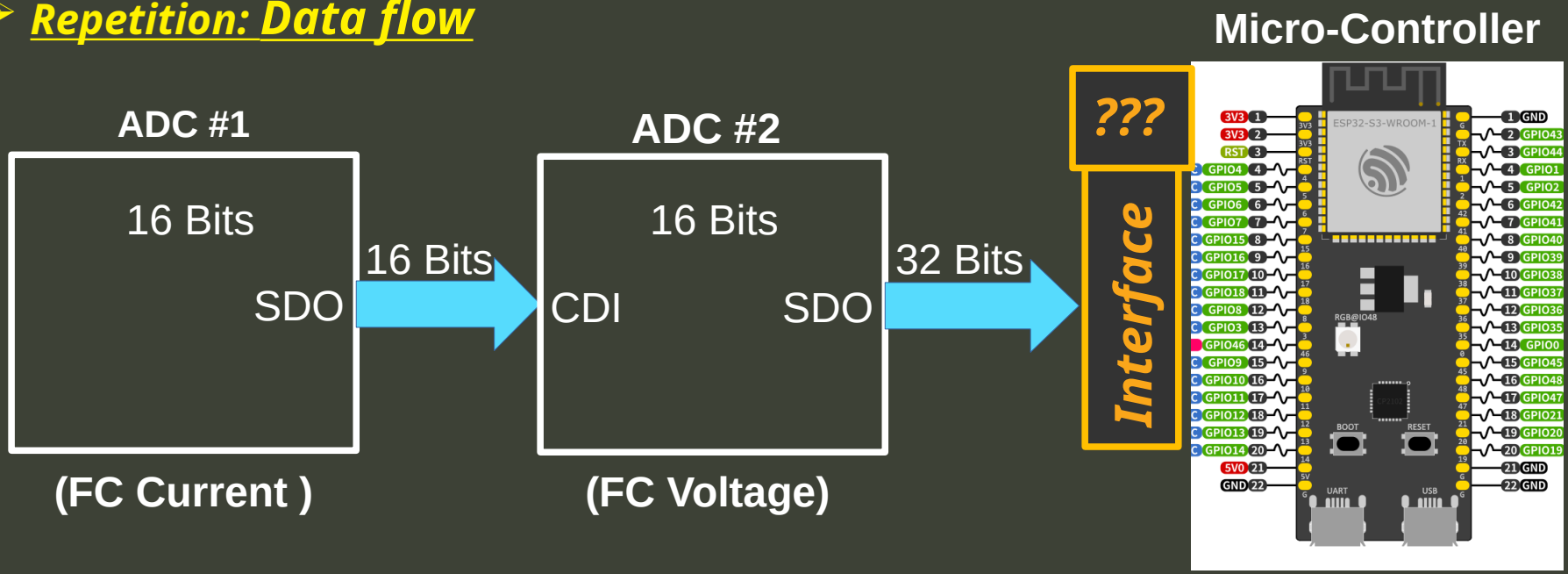
- ***Repetition: ADC chain configuration & Micro-Controller port mapping***



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- ***Repetition: Data flow***



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- ***Repetition: ADS8329 → ADC → data-sheet (extraction):***

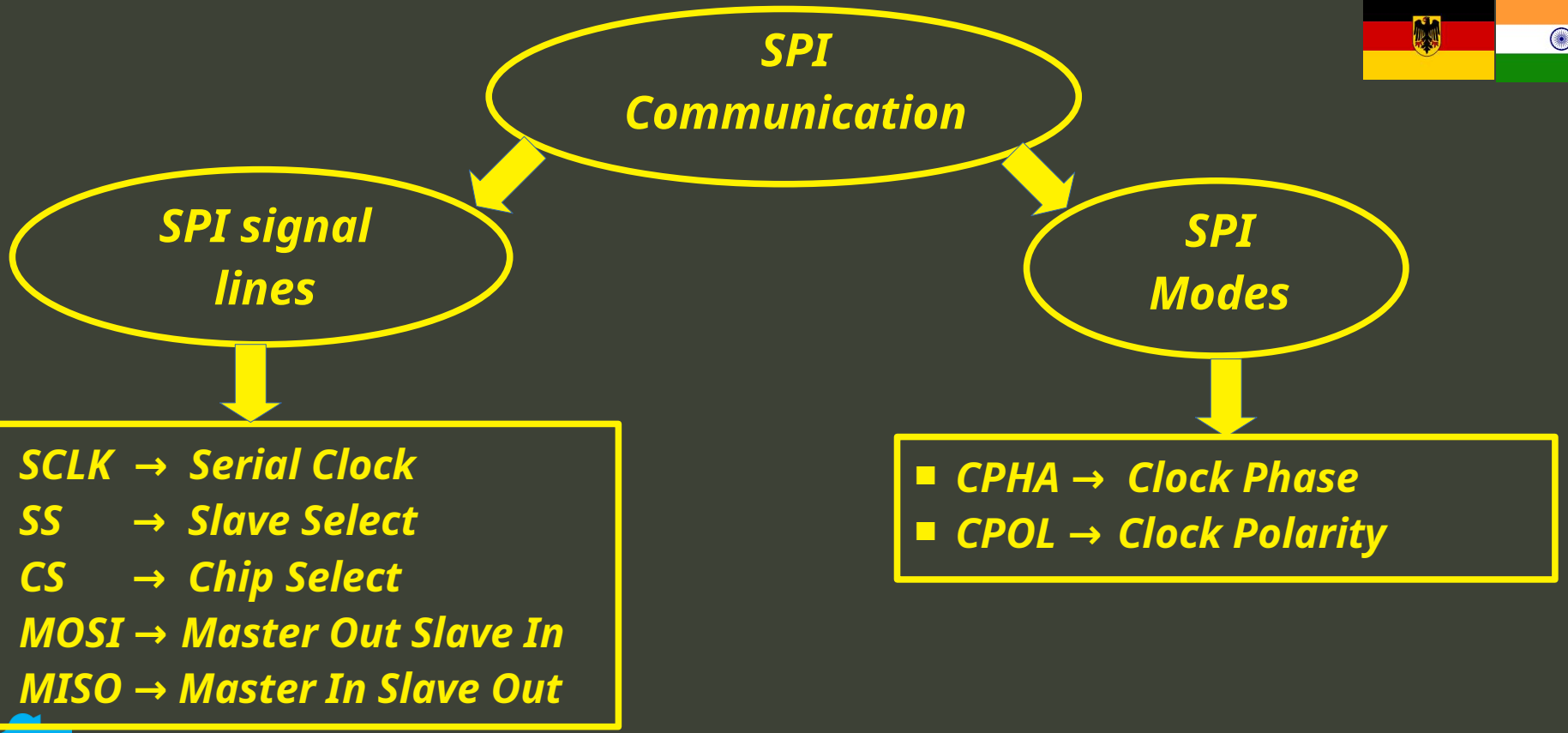
Chain Mode

The ADS8329/30 can operate as a single converter or in a system with multiple converters. System designers can take advantage of the simple high-speed SPI compatible serial interface by cascading them in a single chain when multiple converters are used. A bit in the CFR is used to reconfigure the EOC/$\overline{\text{INT}}$ status pin as a secondary serial data input, chain data input (CDI), for the conversion result from an upstream converter. This is chain mode operation. A typical connection of three converters is shown in Figure 59.

→ SPI = Serial Peripheral Interface

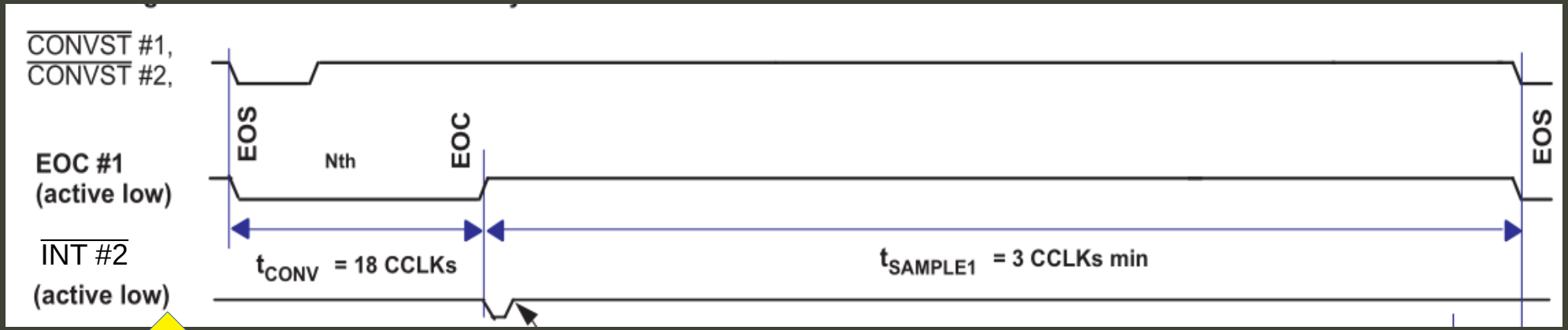
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Project Seminar – Software Dev. - ADC

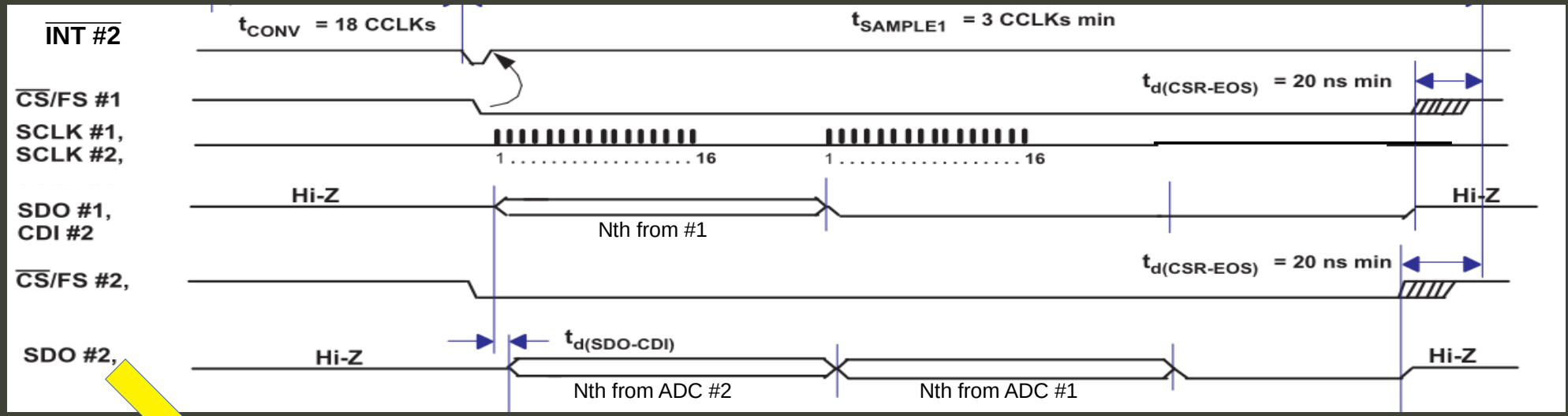
- ***Repetition: ADS8329 → ADC → Timing diagram (1)***



- ***CONVST → Conversion Start***
- ***EOS → End Of Sampling***
- ***EOC → End Of Conversion***
- ***INT → Interrupt***

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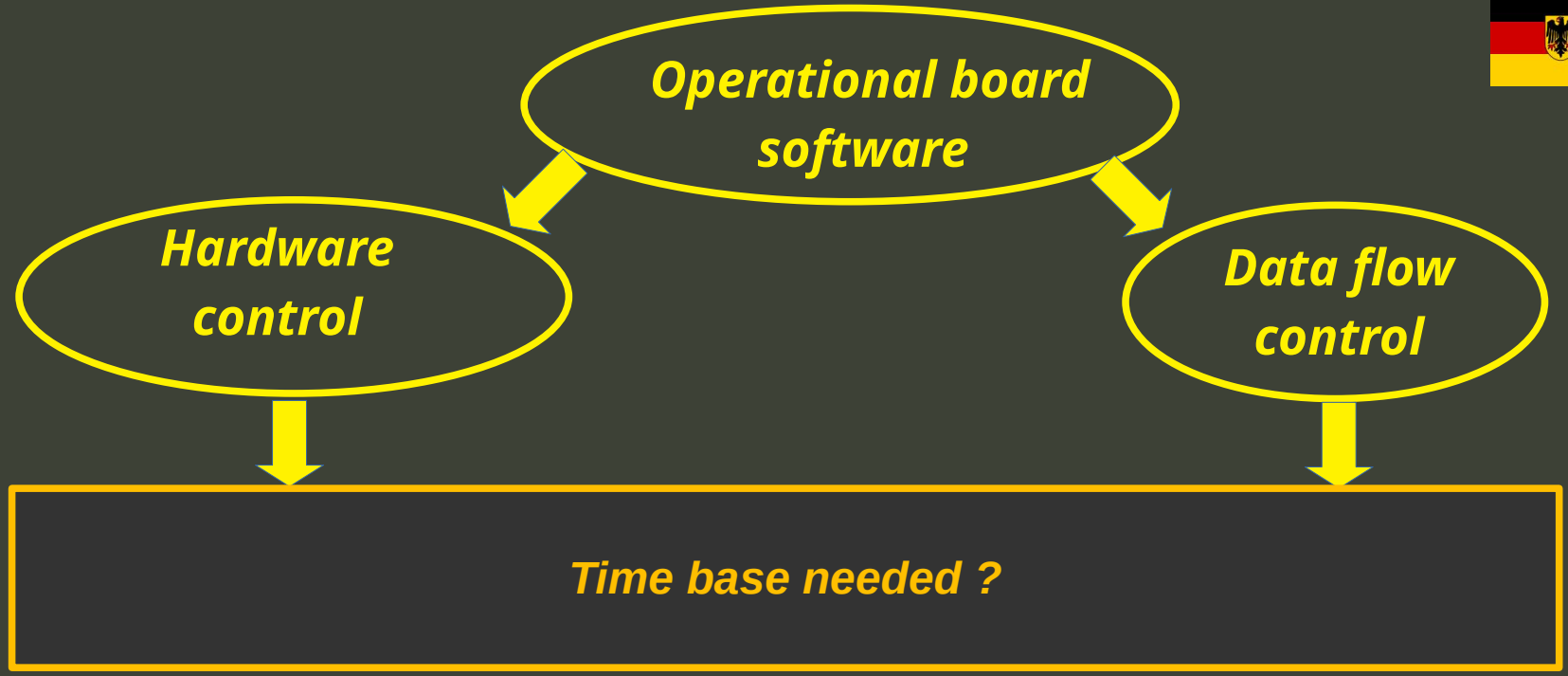
- ***Repetition: ADS8329 → ADC → Timing diagram (2)***



- ***First 16 bit data package belongs to ADC #2***
- ***Second 16 bit data package belongs to ADC #1***

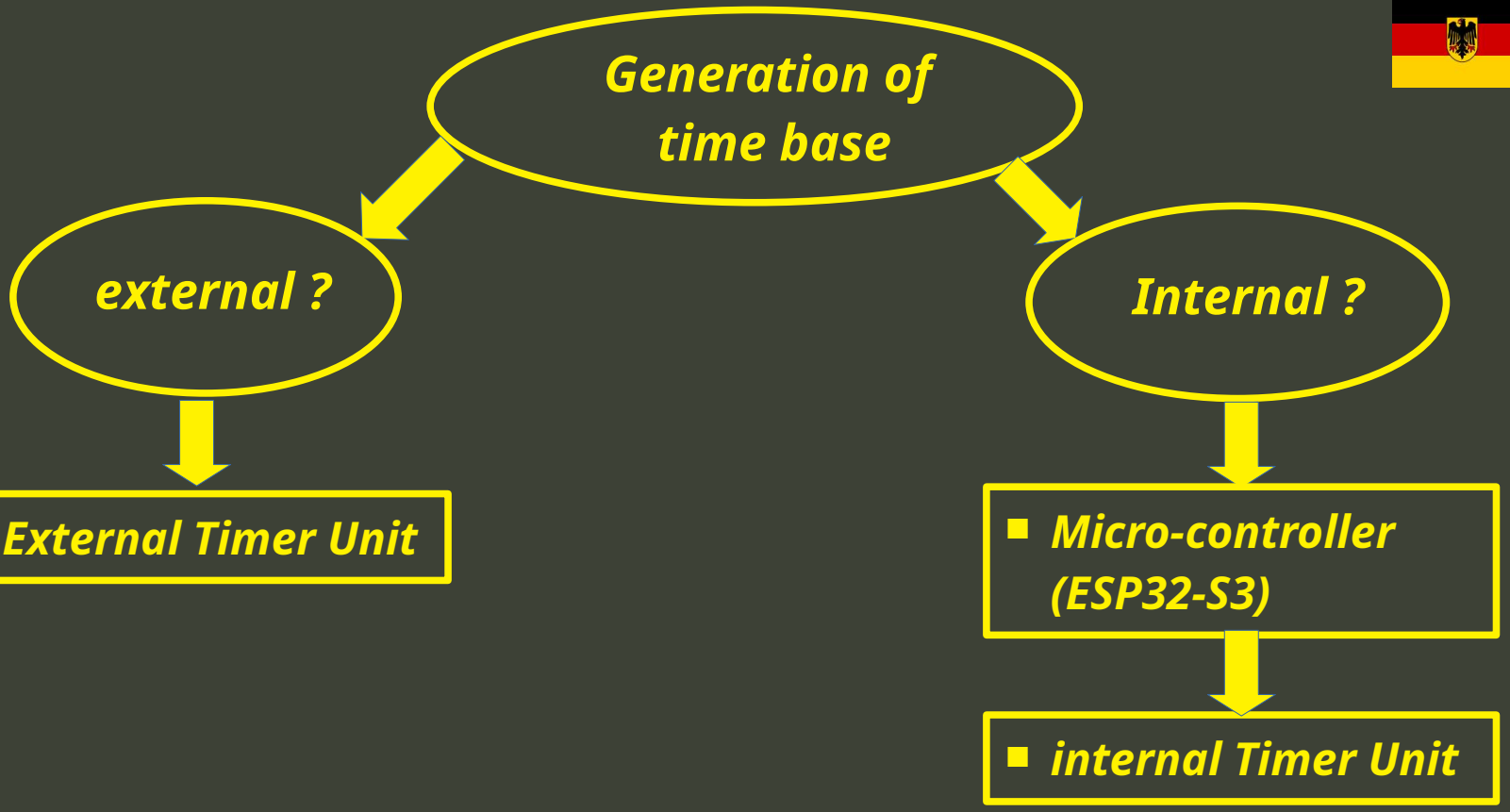
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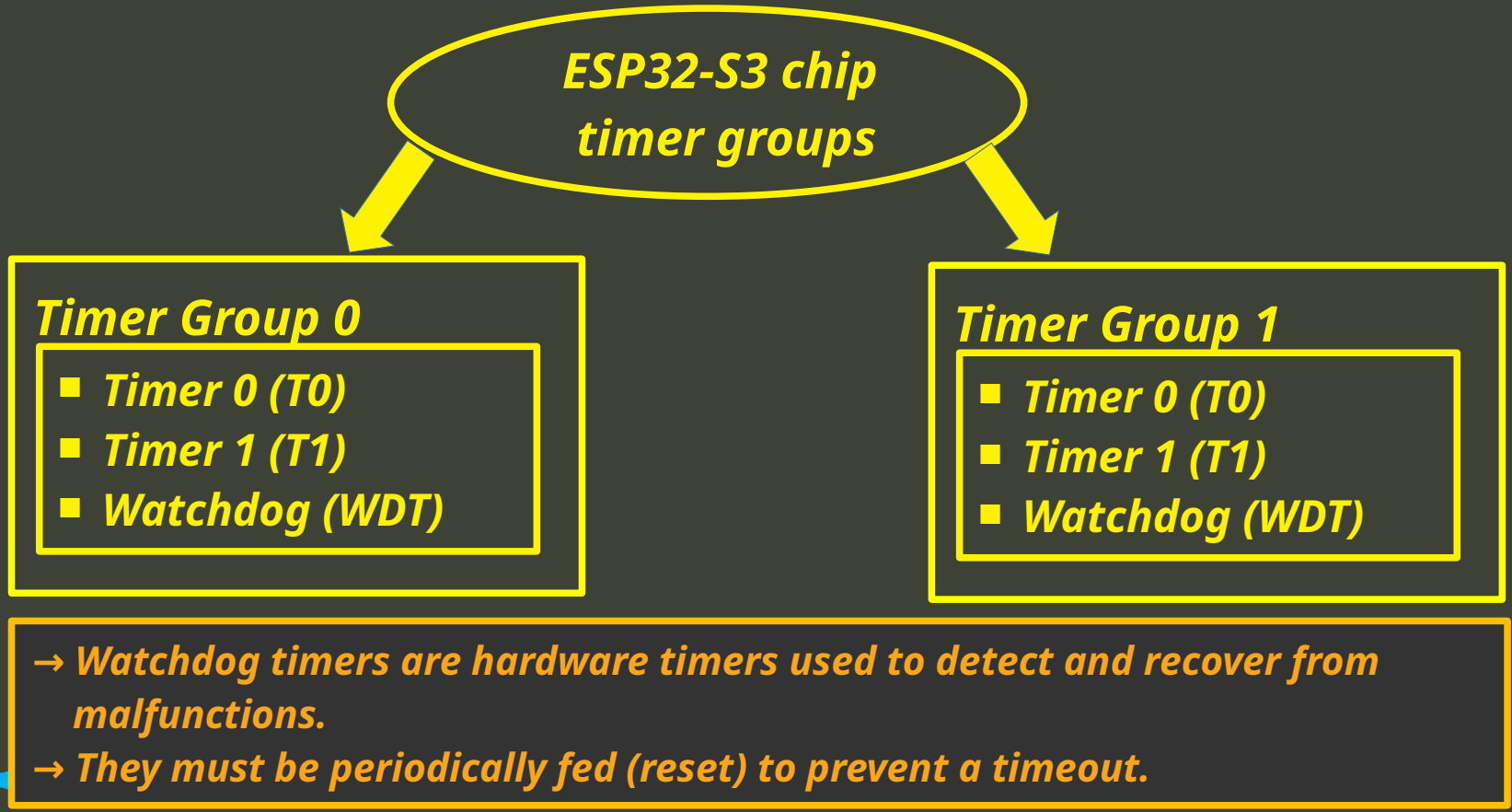


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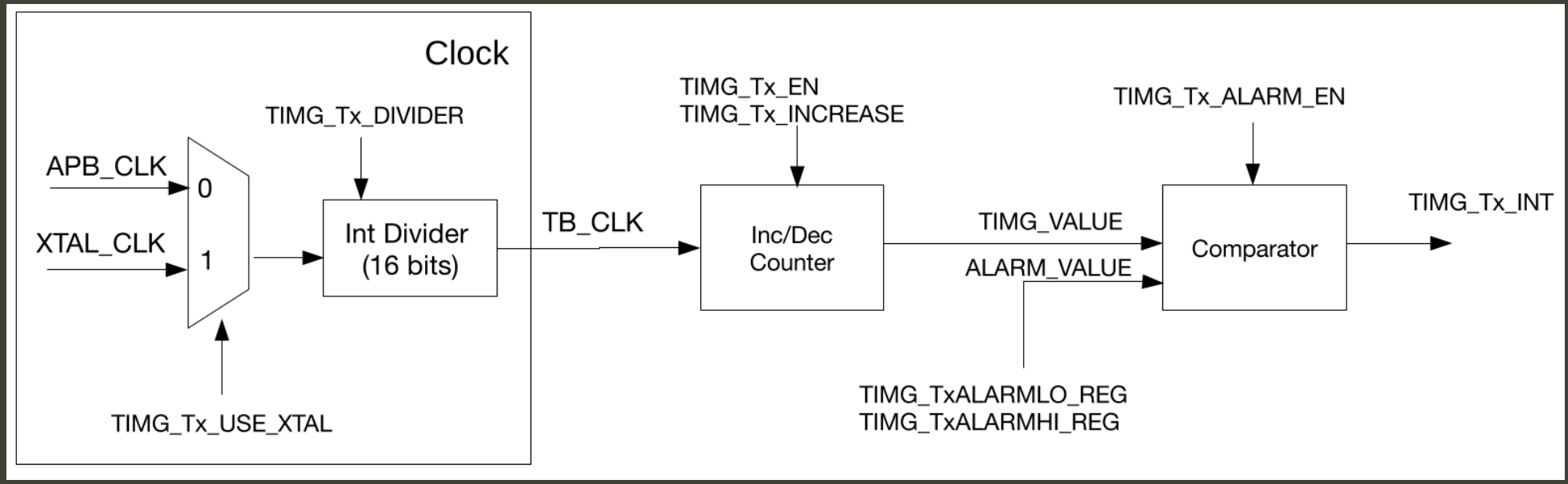
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- ***Timer group architecture: → overview***



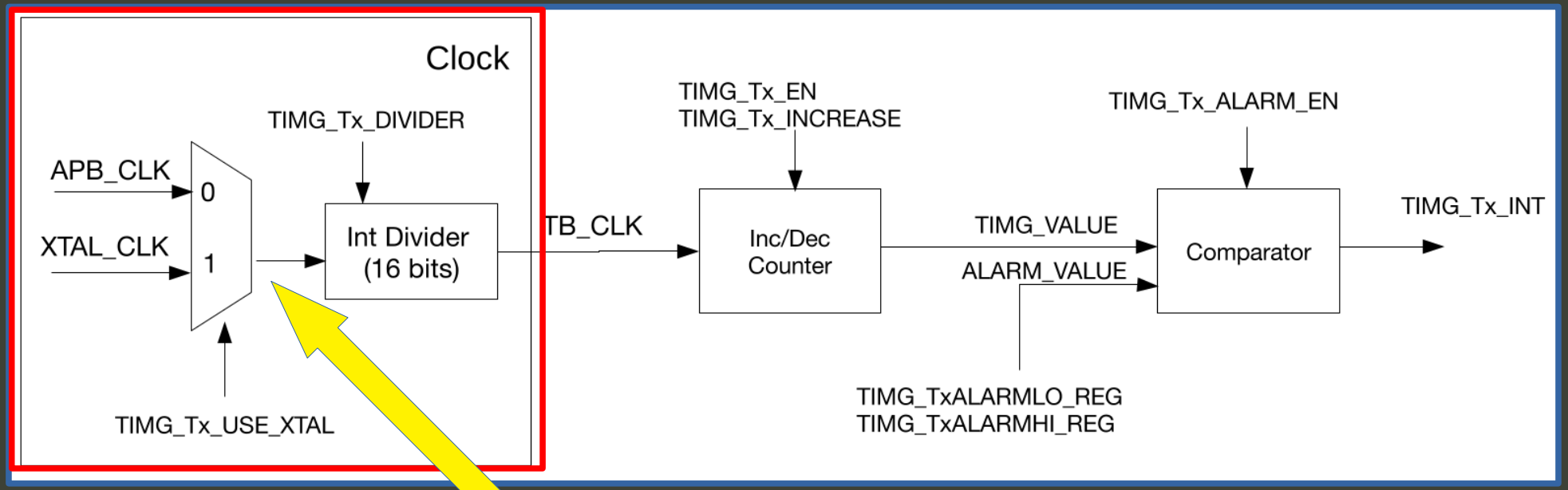
Each timer Tx contains a clock selector, a 16-bit integer divider as a prescaler, a timer-based counter (54-bit) and a comparator for alarm generation.

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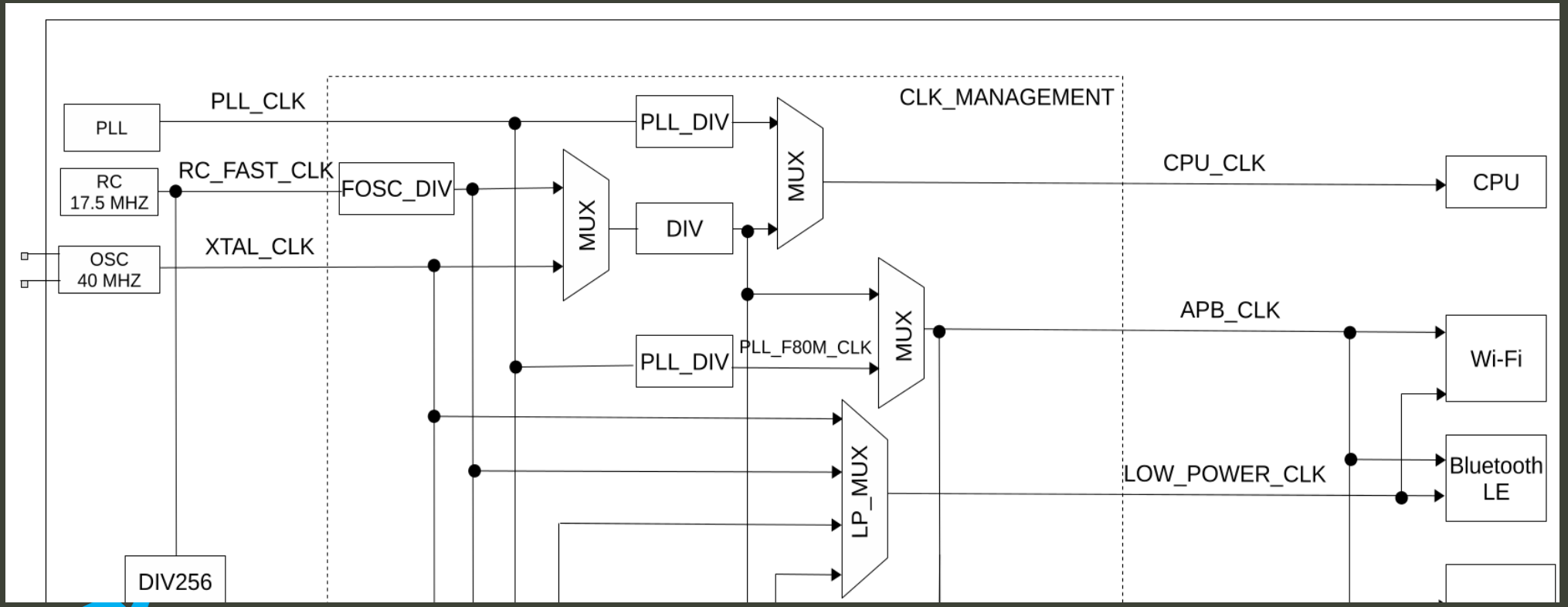
- ***Timer group architecture: → clock source selection***



Each timer can select between the APB clock (APB_CLK) or external clock (XTAL_CLK) as its clock source.
→ We have to read the chapter "Clock" in the technical reference manual!

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- ***Clock structure: → architectural overview***



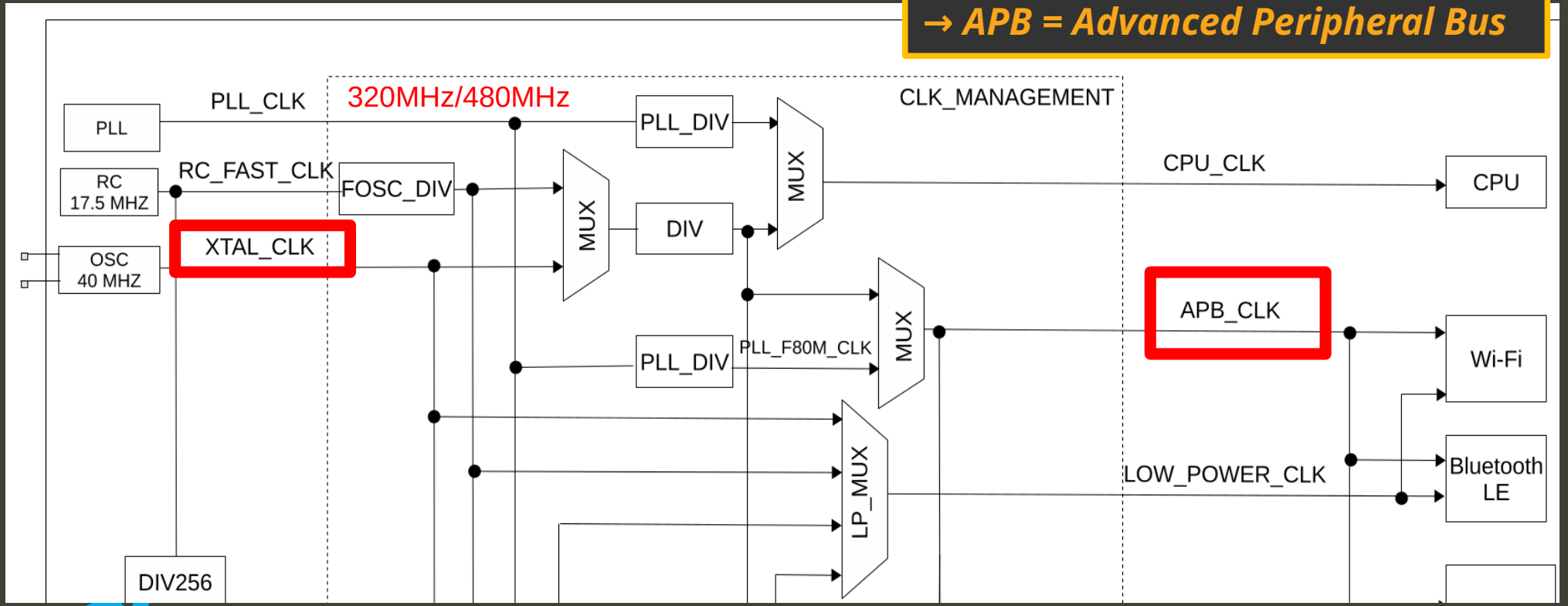
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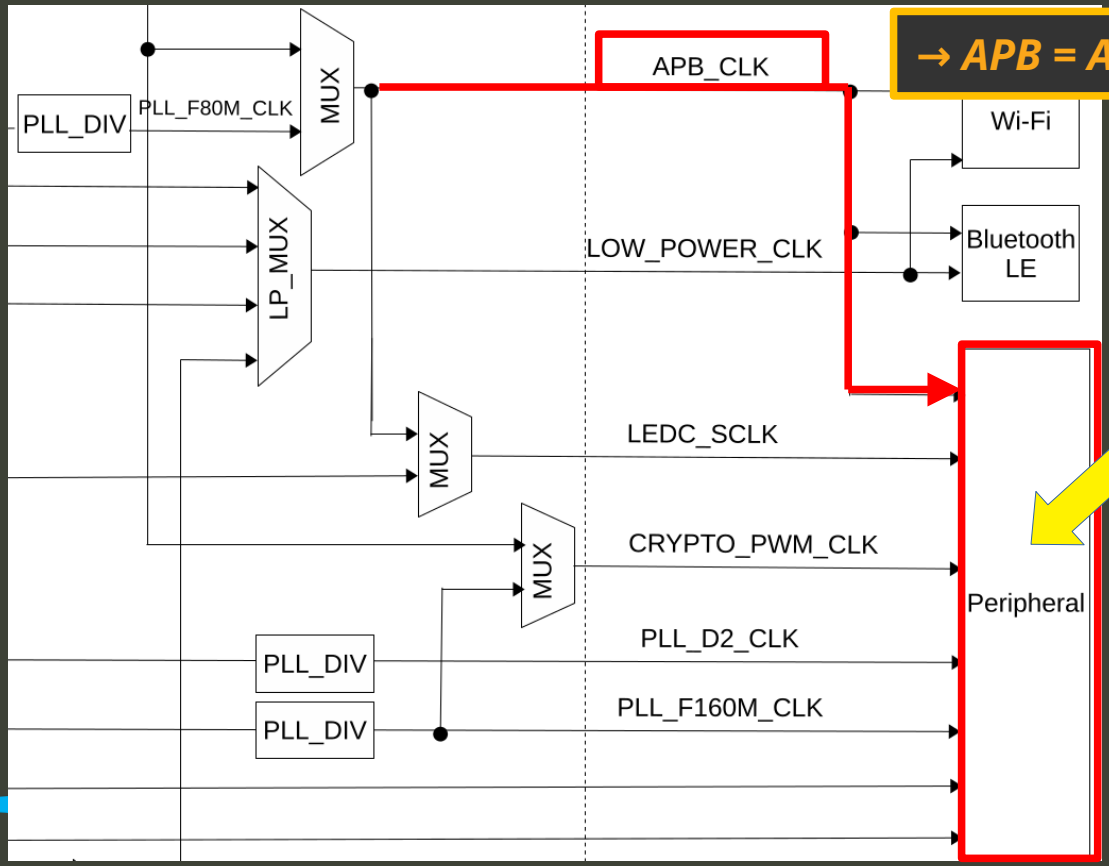
- ***Clock structure: → architectural overview***

→ APB = Advanced Peripheral Bus



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- ***Clock structure → APB-Clock:***

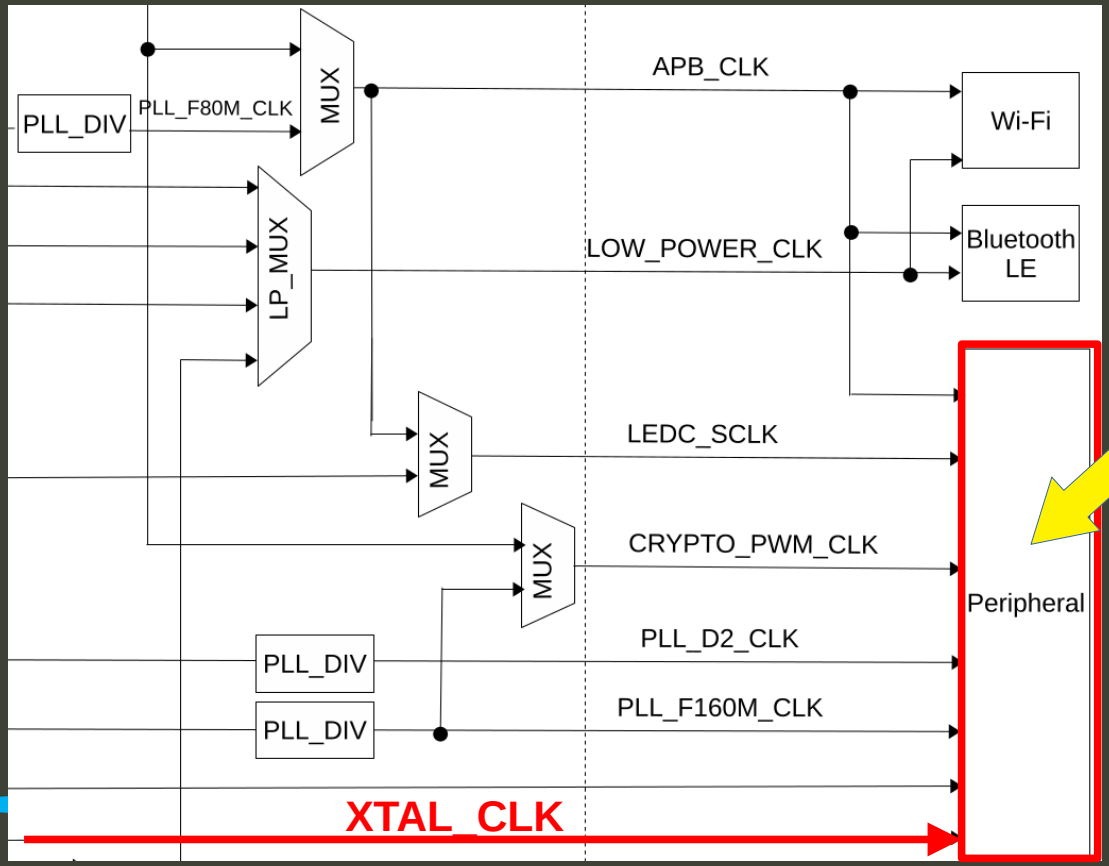


→ APB = Advanced Peripheral Bus

- ***Timer Group 0***
- ***Timer Group 1***

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- ***Clock structure → XTAL-Clock:***

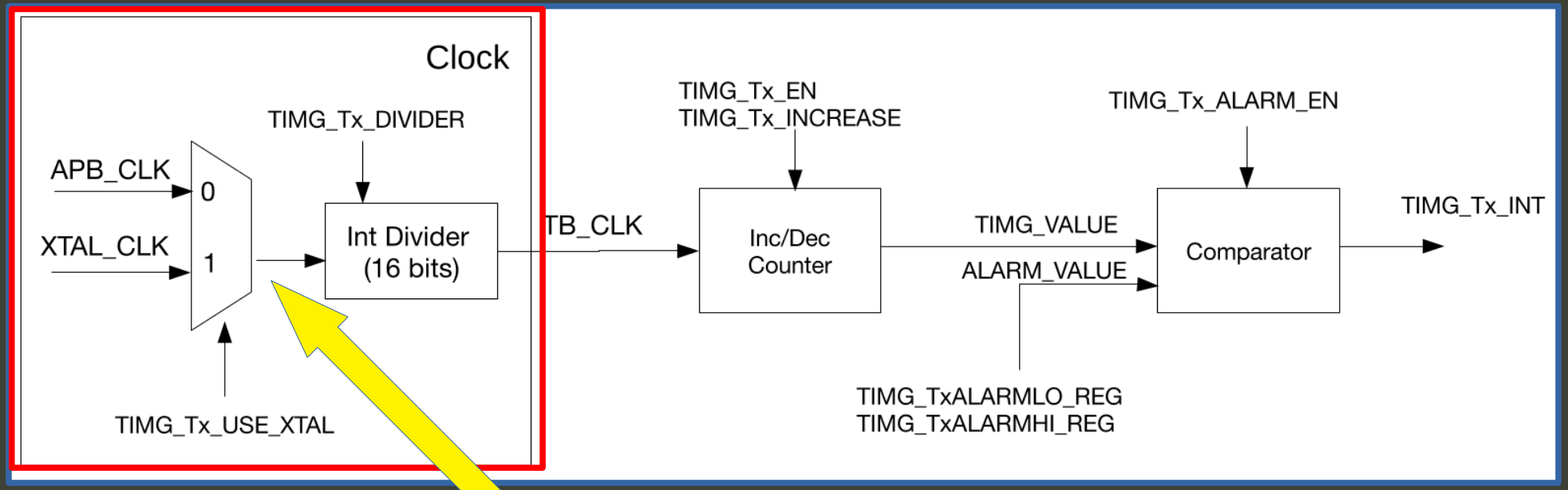


- ***Timer Group 0***
- ***Timer Group 1***

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- ***Timer group architecture: → clock source selection***

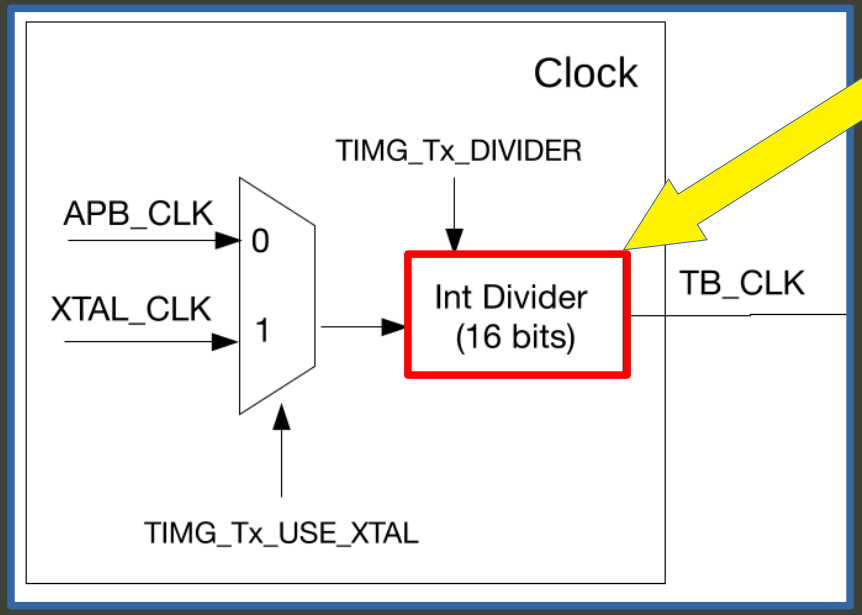


Each timer can select between the APB clock (APB_CLK) or external clock (XTAL_CLK) as its clock source by setting the TIMG_Tx_USE_XTAL field of the TIMG_TxCONFIG_REG register.

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- ***Timer group architecture: → clock divider***

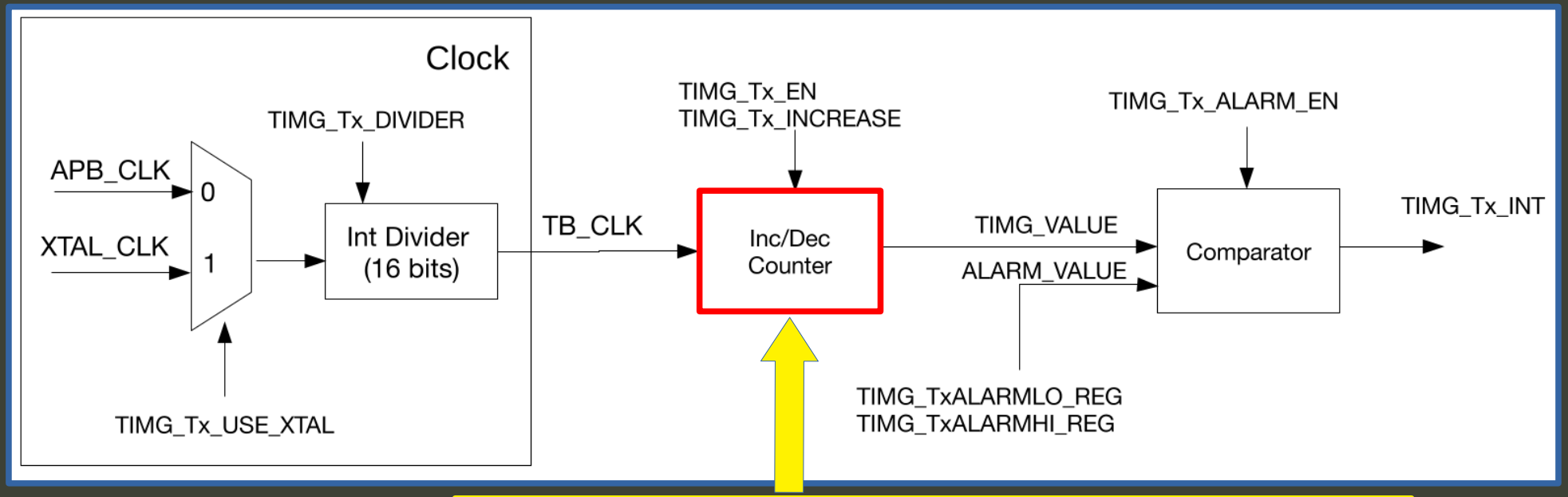


- ***The clock is divided by a 16-bit prescaler to generate the time-base counter clock (TB_CLK).***
- ***When the TIMG_Tx_DIVIDER field is configured as 2 ~ 65536, the divisor of the prescaler would be 2 ~ 65536.***
- ***Programming value 0 to TIMG_Tx_DIVIDER will result in the divisor being 65536.***
- ***Before you modify the 16-bit prescaler, the timer must be disabled (i.e., TIMG_Tx_EN should be cleared).***



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- ***Timer group architecture:***

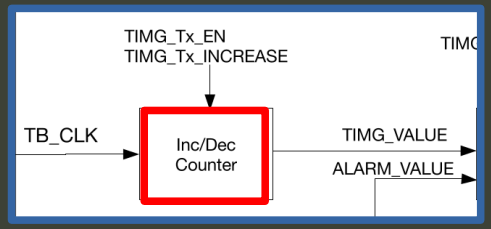


- ***54-bit time-base counter (up/down/up-down)***

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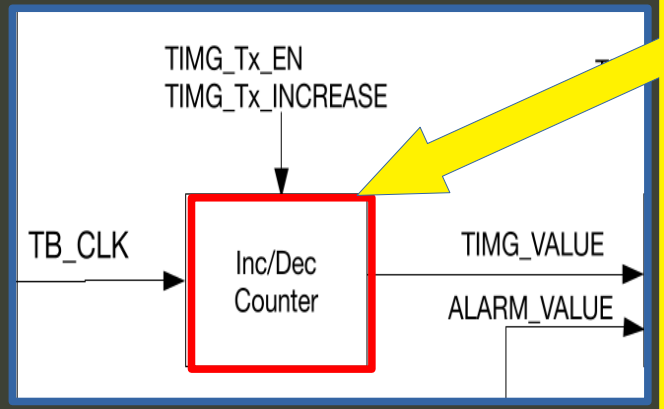
- ***Timer group architecture: → Counter***





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- ***Timer group architecture: → Counter (1)***



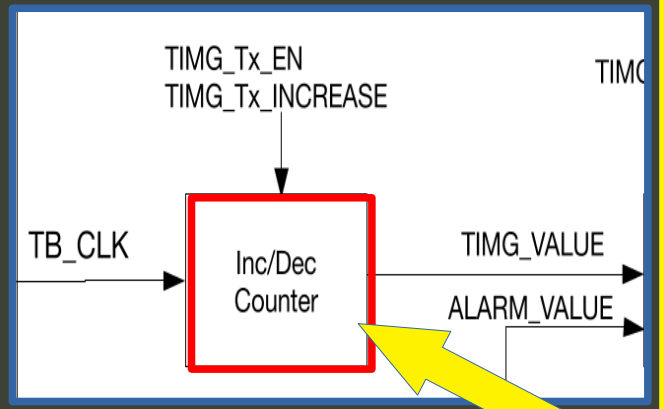
- *TB-counter can be configured to increment or decrement via the TIMG_Tx_INCREASE field.*
- *TB-counter can be enabled or disabled by setting or clearing the TIMG_Tx_EN field, respectively.*
- *When enabled, the time-base counter increments or decrements on each cycle of TB_CLK.*
- *When disabled, the time-base counter is essentially frozen.*

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- ***Timer group architecture: → Counter (2)***

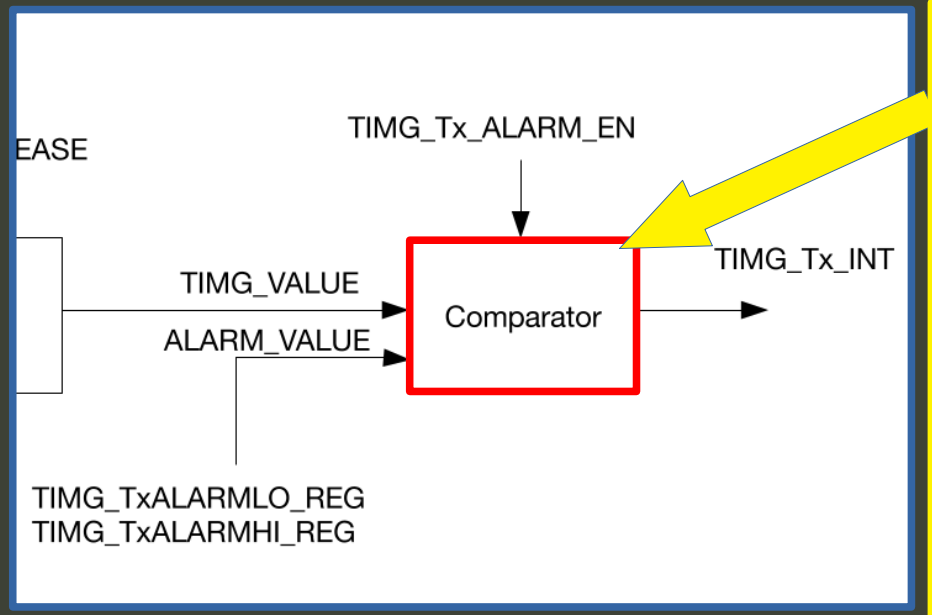


- *By writing any value to the TIMG_TxUPDATE_REG, the current value of the 54-bit timer starts to be latched into the TIMG_TxLO_REG and TIMG_TxHI_REG registers containing the lower 32-bits and higher 22-bits, respectively.*
- *When TIMG_TxUPDATE_REG is cleared by hardware, it indicates the latch operation has been completed and current timer value can be read from the TIMG_TxLO_REG and TIMG_TxHI_REG registers.*
- *TIMG_TxLO_REG and TIMG_TxHI_REG registers will remain unchanged for the CPU to read in its own time until TIMG_TxUPDATE_REG is written to again.*



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- ***Timer group architecture: → Alarm generation***



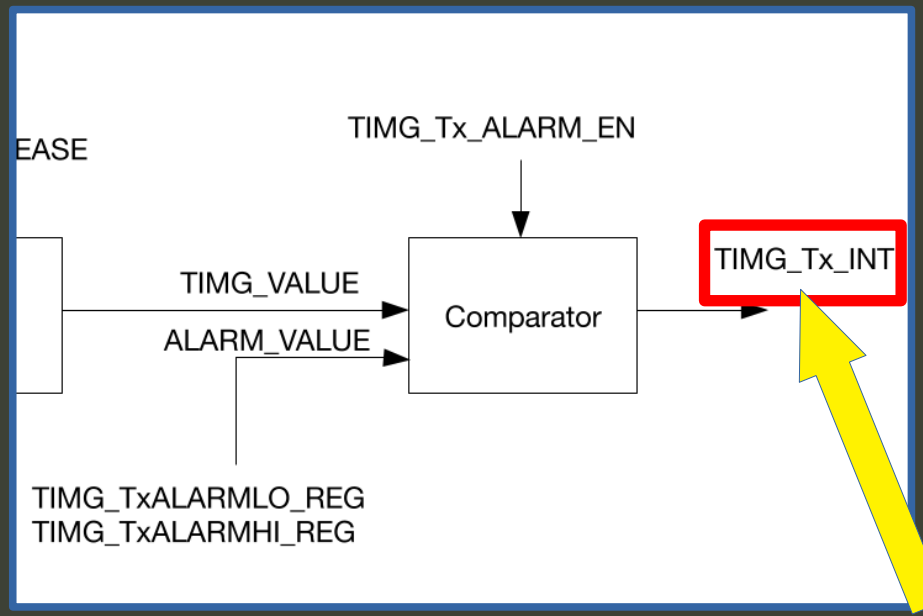
- *A timer can be configured to trigger an alarm when the timer's current value matches the alarm value.*
- *An alarm will cause an interrupt (→ TIMG_Tx_INT) to occur and (optionally) an automatic reload of the timer's current value.*
- *The 54-bit alarm value is configured using TIMG_TxALARMLO_REG and TIMG_TxALARMHI_REG*
- *The alarm is enabled by setting the TIMG_Tx_ALARM_EN field.*

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- ***Timer group architecture: → Alarm generation → Interrupts (1)***

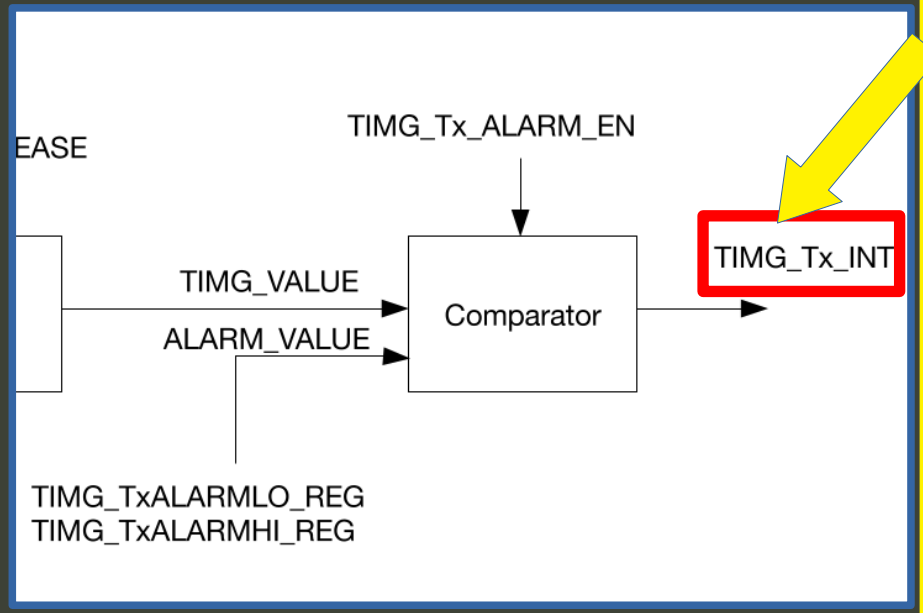


- *Each timer has its own interrupt line that can be routed to the CPU, and thus each timer group has a total of three interrupt lines.*
- *Timers generate level interrupts that must be explicitly cleared by the CPU on each triggering.*
- *Interrupts are triggered after an alarm occurs.*
- *Level interrupts will be held high after an alarm occurs, and will remain so until manually cleared.*



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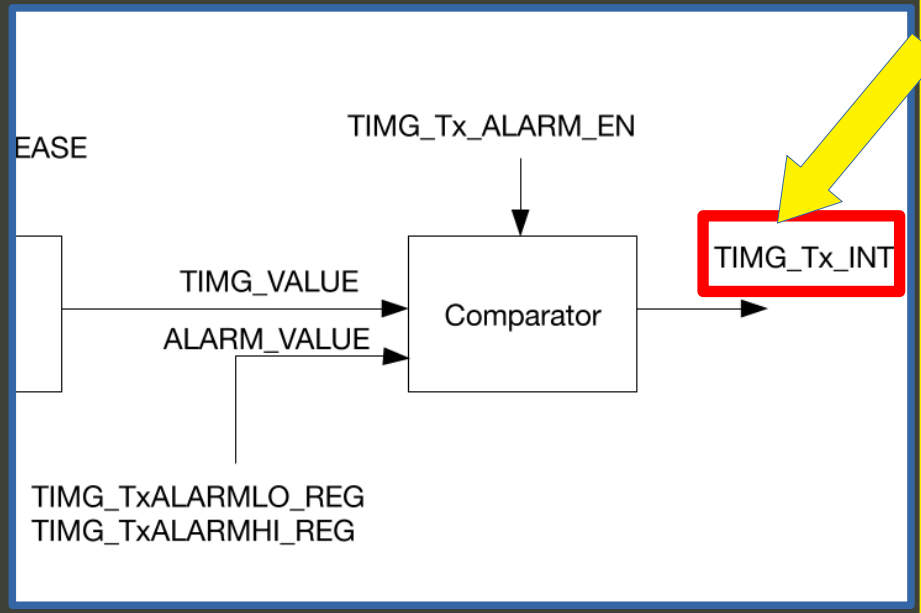
- ***Timer group architecture: → Alarm generation → Interrupts (2)***



- *To enable a timer's interrupt, the TIMG_Tx_INT_ENA bit should be set.*
- *The interrupts of each timer group are governed by a set of registers.*
- *TIMG_Tx_INT_RAW :*
  → An alarm event sets it to 1. The bit will remain set until the timer's corresponding bit in TIMG_Tx_INT_CLR is written.
- *TIMG_Tx_INT_ST : → Reflects the status of each timer's interrupt and is generated by masking the bits of TIMG_Tx_INT_RAW with TIMG_Tx_INT_ENA.*

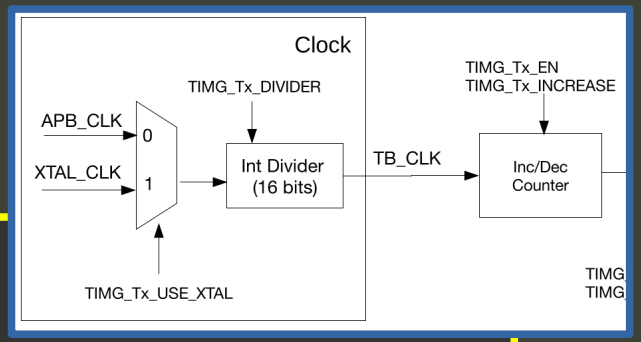
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- ***Timer group architecture: → Alarm generation → Interrupts (3)***



- *TIMG_Tx_INT_ENA : → Used to enable or mask the interrupt status bits of timers within the group.*
- *TIMG_Tx_INT_CLR : → Used to clear a timer's interrupt by setting its corresponding bit to 1. The timer's corresponding bit in TIMG_Tx_INT_RAW and TIMG_Tx_INT_ST will be cleared as a result.*
- *Note that a timer's interrupt must be cleared before the next interrupt occurs !!!*

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- ***Exercise 11: Configuration Timer as Periodic Alarm (1)***

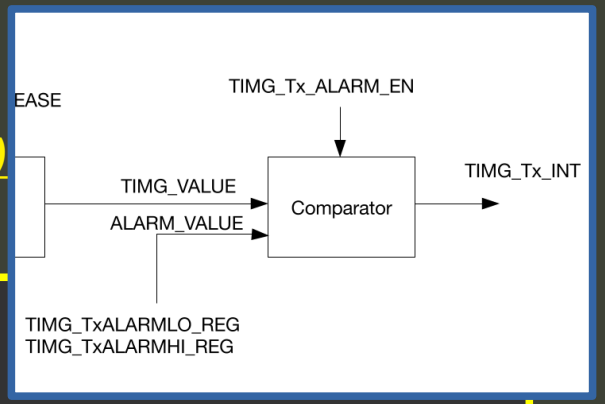
1. Configure the time-base counter

- Select clock source by setting or clearing TIMG_Tx_USE_XTAL field.
- Configure the 16-bit prescaler by setting TIMG_Tx_DIVIDER.
- Configure the timer direction by setting or clearing TIMG_Tx_INCREASE.
- Set the timer's starting value by writing the starting value to TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI, then reloading it into the timer by writing any value to TIMG_TxLOAD_REG.

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- ***Exercise 11: Configuration Timer as Periodic Alarm (2)***



2. Configure the alarm

- Configure the alarm value by setting TIMG_TxALARMLO_REG and TIMG_TxALARMHI_REG.
- Enable interrupt by setting TIMG_Tx_INT_ENA.

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- ***Exercise 11: Configuration Timer as Periodic Alarm (3)***

3. Enable auto reload

- by setting TIMG_Tx_AUTORELOAD and configure the reload value via TIMG_Tx_LOAD_LO and TIMG_Tx_LOAD_HI.

4. Start the alarm

- by setting TIMG_Tx_ALARM_EN.



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- ***Exercise 11: Configuration Timer as Periodic Alarm (4)***

5. Handle the alarm interrupt (repeat on each alarm iteration). [ISR]

- Clear the interrupt by setting the timer's corresponding bit in TIMG_Tx_INT_CLR.
- If the next alarm requires a new alarm value and reload value (i.e., different alarm interval per iteration), then TIMG_TxALARMLO_REG, TIMG_TxALARMHI_REG, TIMG_Tx_LOAD_LO, and TIMG_Tx_LOAD_HI should be reconfigured as needed.
- Otherwise, the aforementioned registers should remain unchanged.



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- ***Exercise 11: Configuration Timer as Periodic Alarm (5)***

6. Stop the timer (on final alarm iteration).

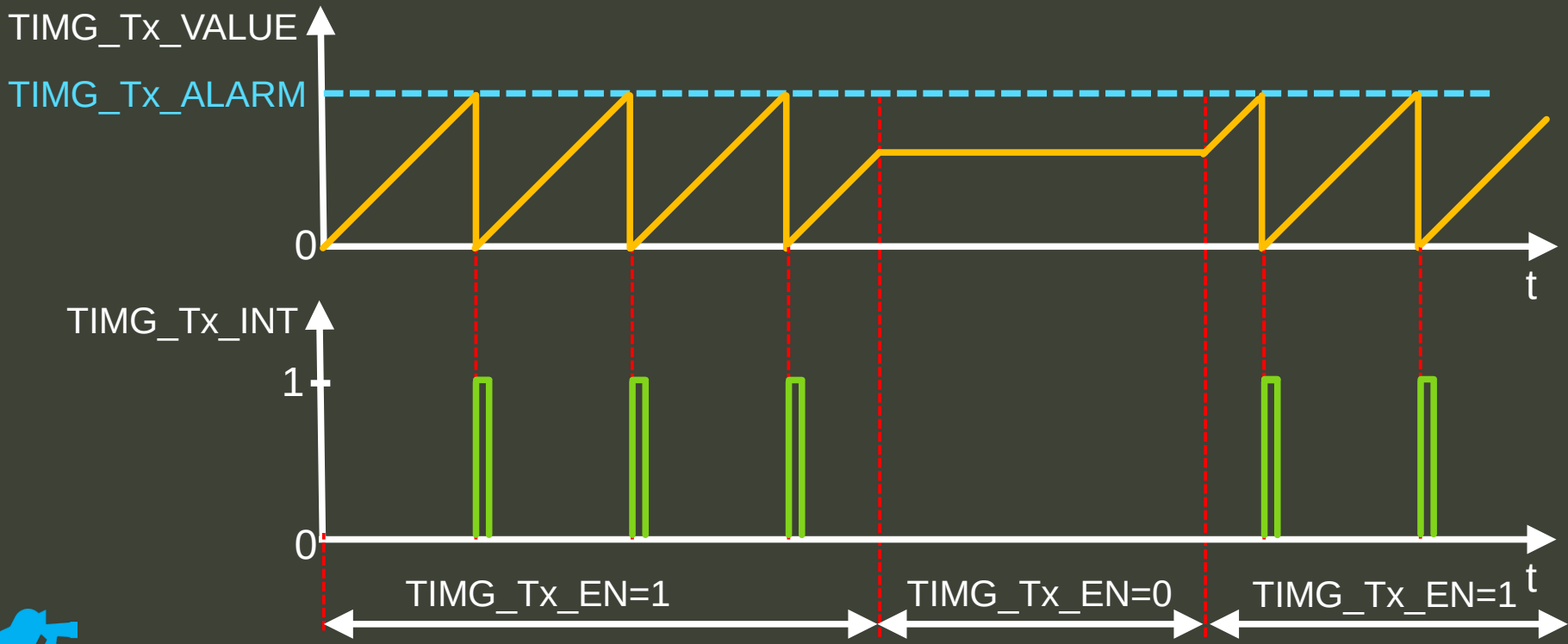
- Clear the interrupt by setting the timer's corresponding bit in TIMG_Tx_INT_CLR.
- Disable the timer by clearing TIMG_Tx_EN.



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- ***Exercise 11: Configuration Timer as Periodic Alarm (6)***



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- ***Exercise 12: Configuration Timer as Periodic Alarm → Calculations***

Given:

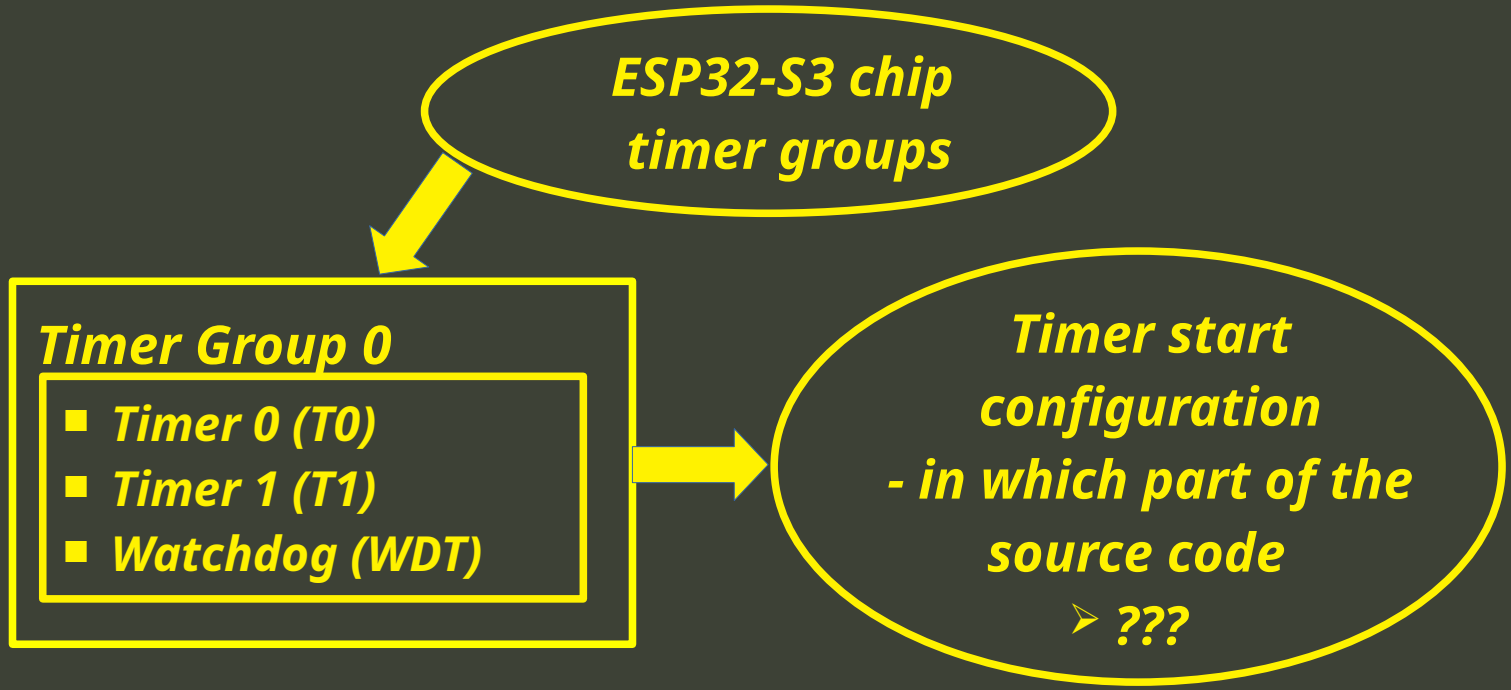
- Use from Timer group 0 timer T0 as periodic alarm.
- TIMG_T0_EN=1
- External clock (XTAL_CLK) frequency = 40MHz
- TIMG_T0_DIVIDER=0x000A
- TIMG_T0ALARMLO_REG= 0x00002710
- TIMG_T0ALARMHI_REG= 0

→ Exercise 12 is presented in the lecture!

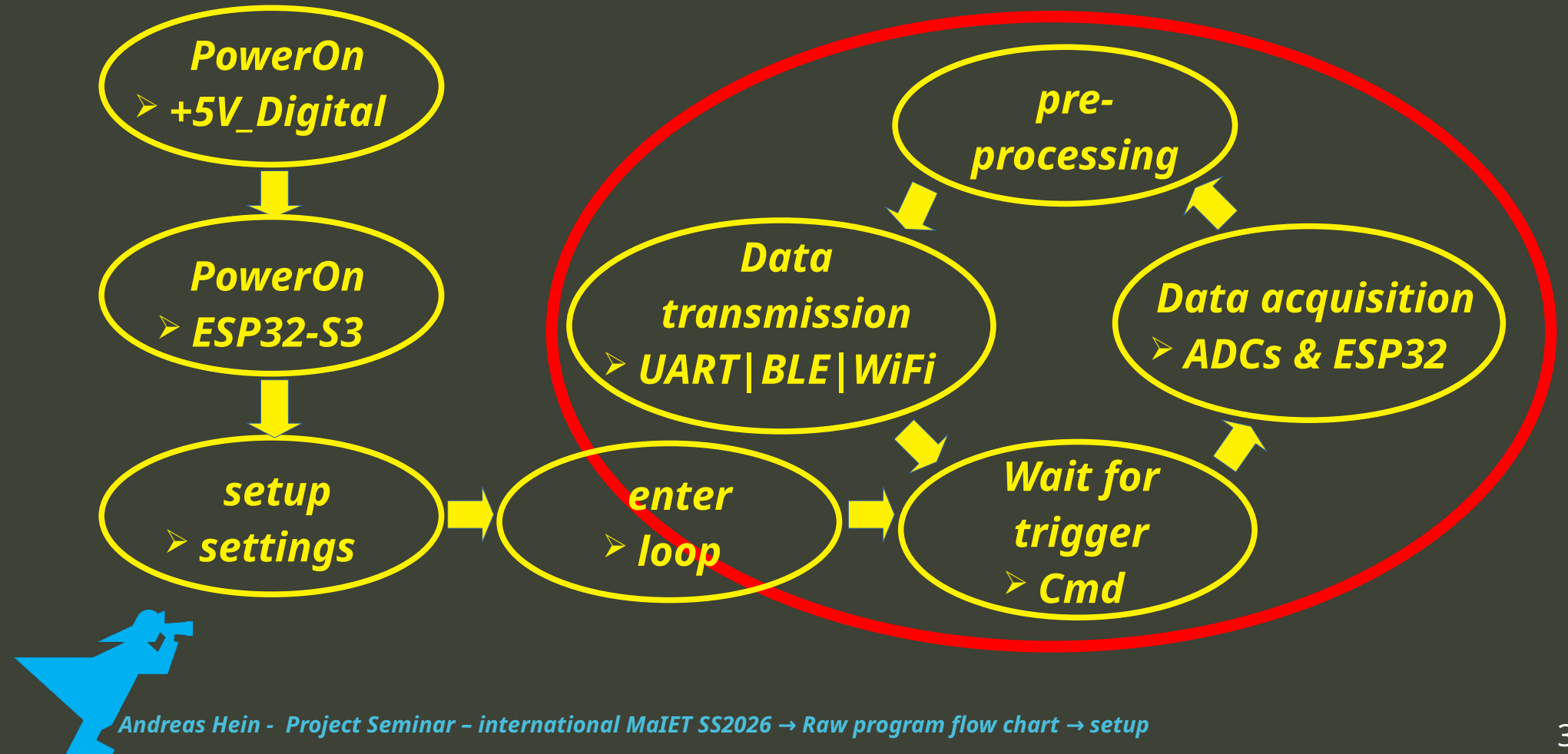
Wanted:

- a) Frequency of time-base counter clock (TB_CLK)
- b) Period of TIMG_T0_INT

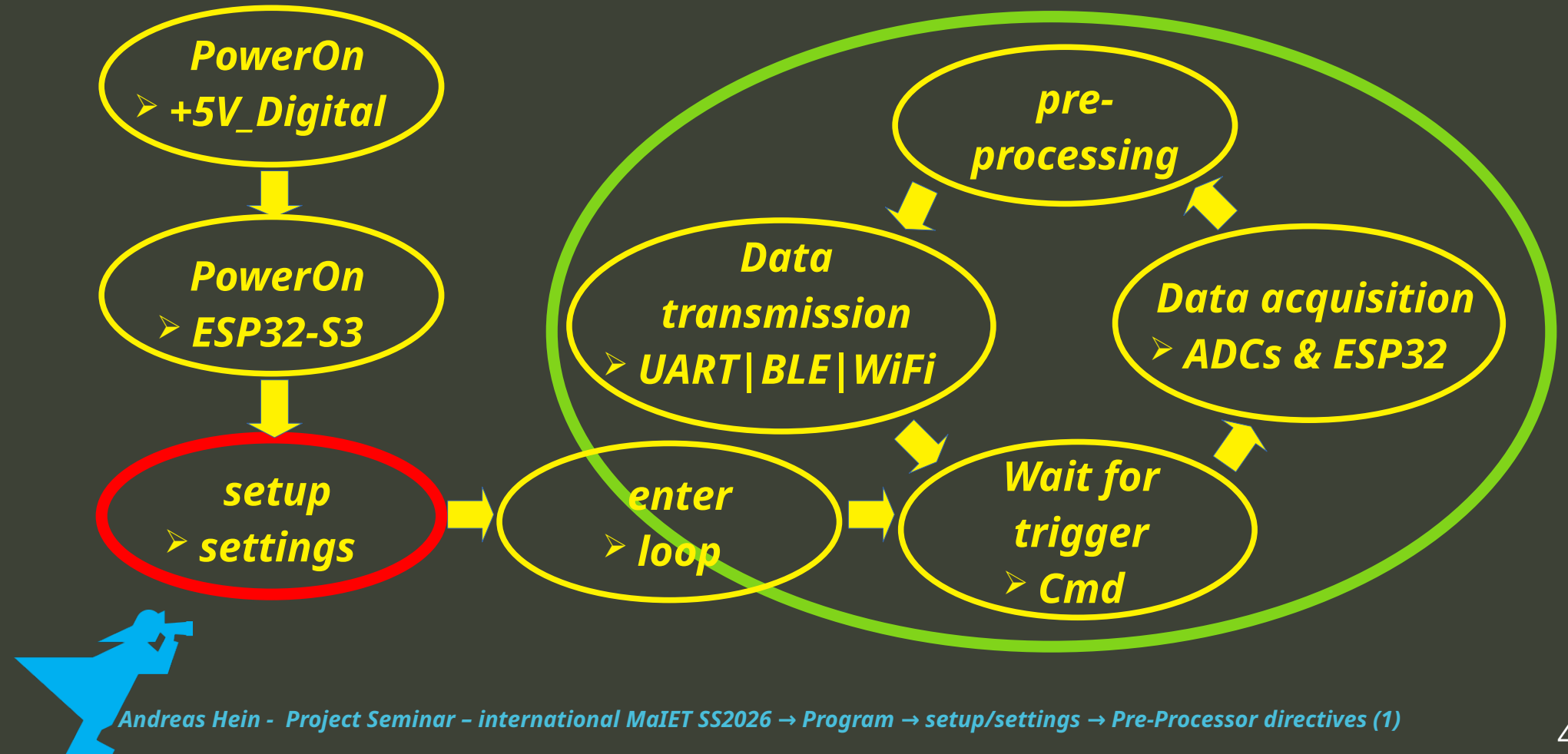
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Pre-processor directives (1)

Include-statements:

- *#include <Arduino.h>*
- *#include <SPI.h>*
- *#include <stdlib.h>*
- *...*

Definitions:

- *#define SPI_DATAMODE 1*
- *#define BUSY_PIN GPIO_NUM_9*
- *#define CNV_PIN GPIO_NUM_14*
- *#define SCLK_PIN GPIO_NUM_12*

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Pre-processor directives (2)

Definitions:

- *#define MISO_PIN GPIO_NUM_13*
- *#define MOSI_PIN GPIO_NUM_11*
- *#define SS_PIN GPIO_NUM_10*
- *#define SPI_CLK 4000000*
- *#define NUM_SAMPLES_MEM 20000*
- *...*

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Declarations & Definitions of functions

Declarations of functions:

- *void spiCom32(SPIClass *spi, uint32_t data, uint8_t spi_datamode);*
- *...*

Definitions of functions:

- *ISR to hamdle TIMG_T0_INT interrupt event*
- *ISR to handle GPIO_BUSY interrupt event*
- *Data transfer via SPI → function: spiCom32*
- *Data transfer via BLE → function: bleCom*

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Declarations & Definitions of variables

→ *hw_timer_t *Timer0_Cfg = NULL;*
→ *uint32_t spi_dummy_data = 0x34BB12A0;*
→ *uint32_t receivedRawValue = 0;*
→ *uint8_t cnt_tmp_samples = 0;*
→ *uint16_t meas_fuel_cell_voltage_samples[NUM_SAMPLES_MEM]={0};*
→ *uint16_t meas_fuel_cell_current_samples[NUM_SAMPLES_MEM]={0};*

→ *Timer 0 alarm value*
→ *...*

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Definition/Configuration of ports (GPIOs)

→ *Pin-Mode: input or output*
→ *Pin-Init level: HIGH or LOW*
→ *Pin-Interrupt-Mode: rising/falling edge*

→ *CNV_GPIO14: OUTPUT*
→ *BUSY_GPIO9: INPUT → interrupt sensitivity → falling edge*
→ *ADC1_CS_GPIO17: OUTPUT*
→ *ADC2_CS_GPIO18: OUTPUT*
→ *...*

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Setup of clock-system and peripheral units

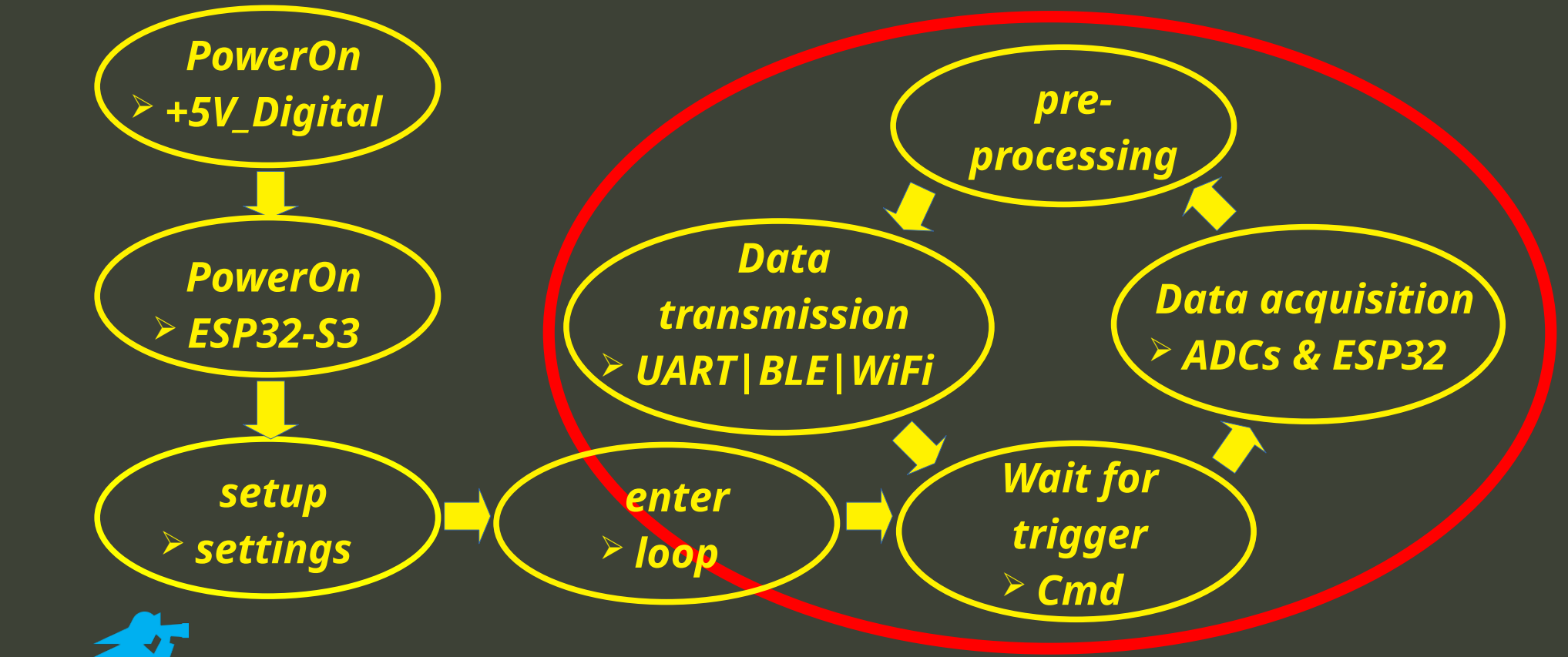
→ *Setup and Configuration of Clock system: CPU-Clock, XTAL_CLK, APB_CLK*

→ *Setup and Configuration of Timer 0: Periodic Alarm Mode*

→ *Setup and Configuration of SPI: SPICLK_SPEED, SPI_MODE*

→ ...

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Loop: Program flow statements

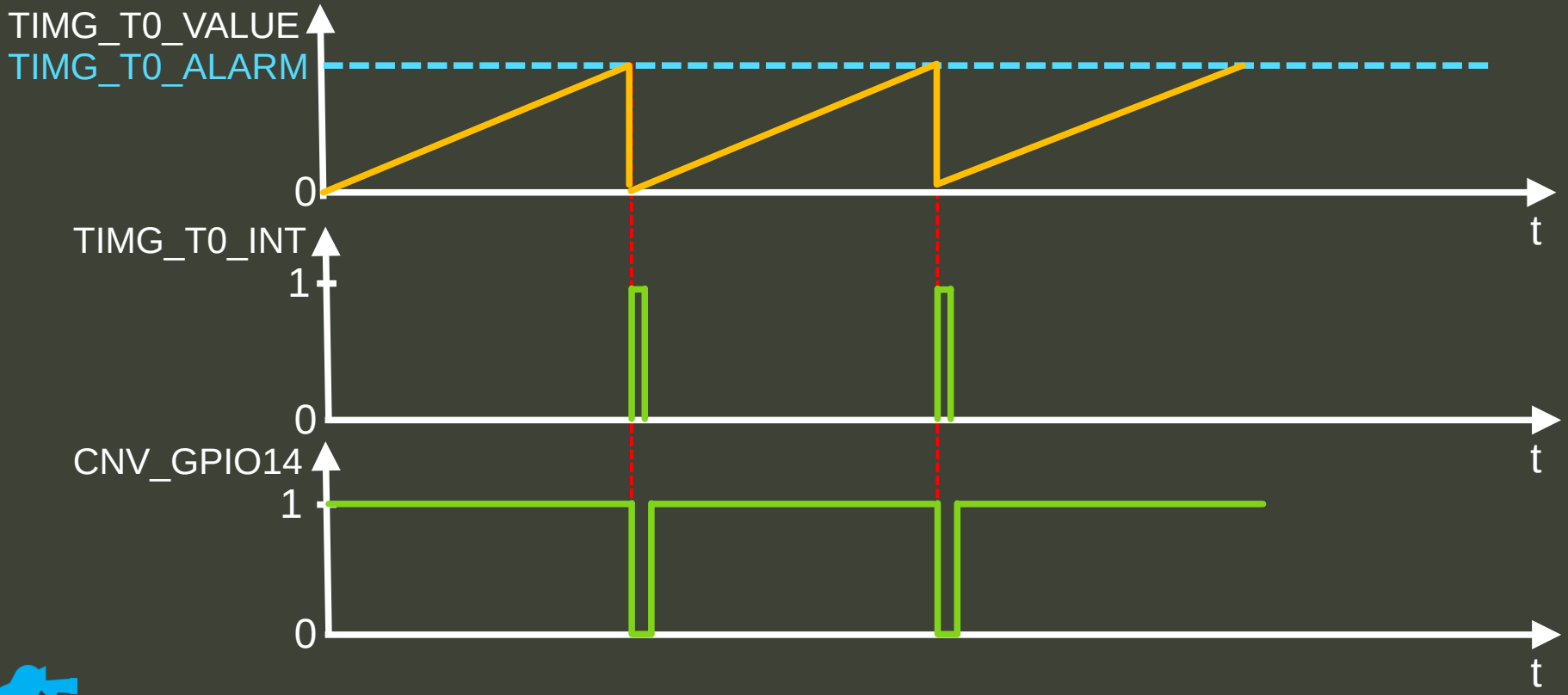
1. Timer 0 of Timer Group 0 ENABLED
2. Timer 0 TIMG_INT interrupt event occurs
3. Timer 0 ISR: CNV_GPIO14 level change from HIGH to LOW to HIGH
4. Timer 0 TIMG_INT cleared
5. ADC1 & ADC2: Analog to Digital Conversion running
6. EOC event of ADC1: Causes falling edge on INT (GPIO9 BUSY)
7. GPIO-interrupt occurs on GPIO9_BUSY
8. SPI-communication: Transmission of conversion results from ADCs to ESP32-S3
9. Data transmission via BLE
10. repeat from step 2

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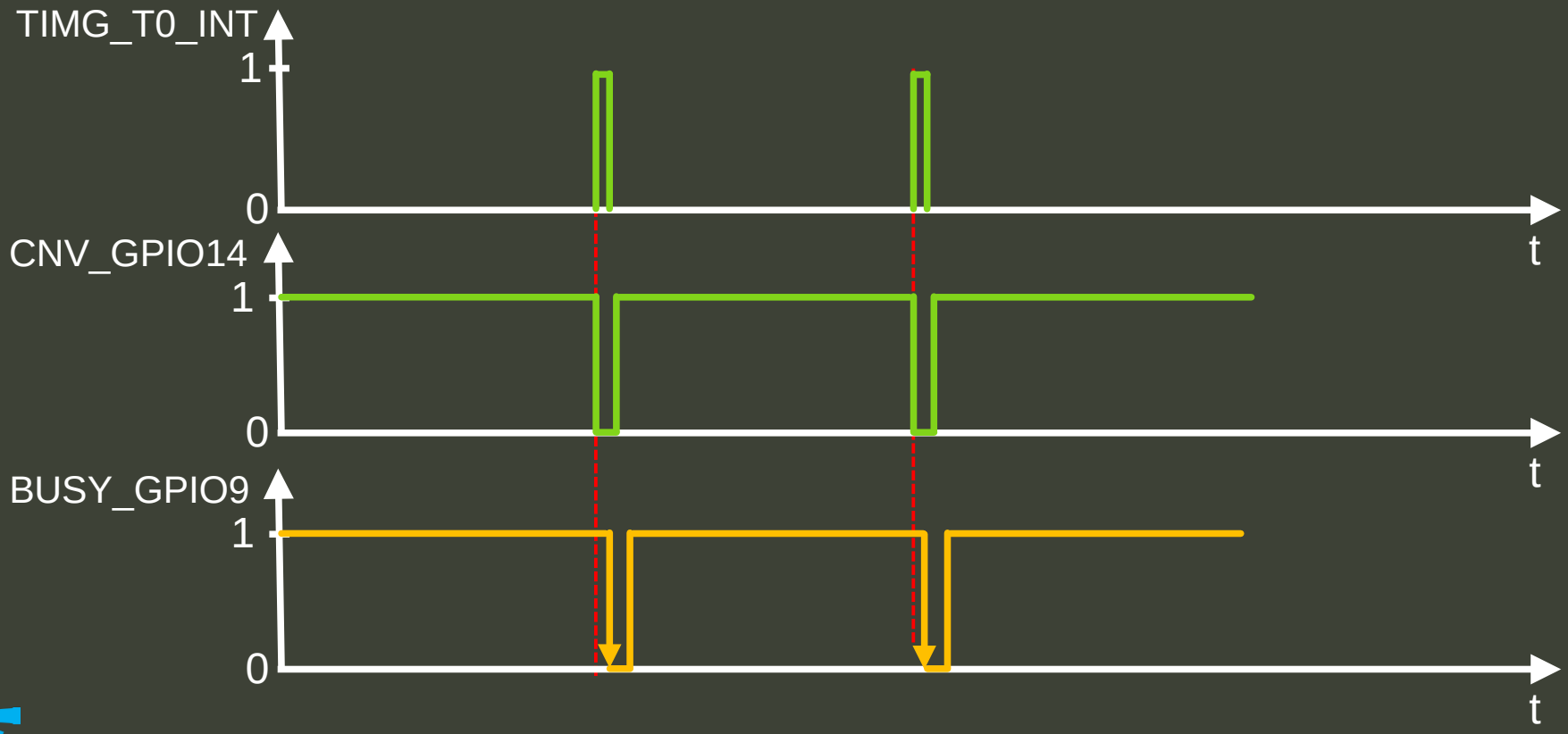
- ***Loop: Program flow timing diagram (1)***





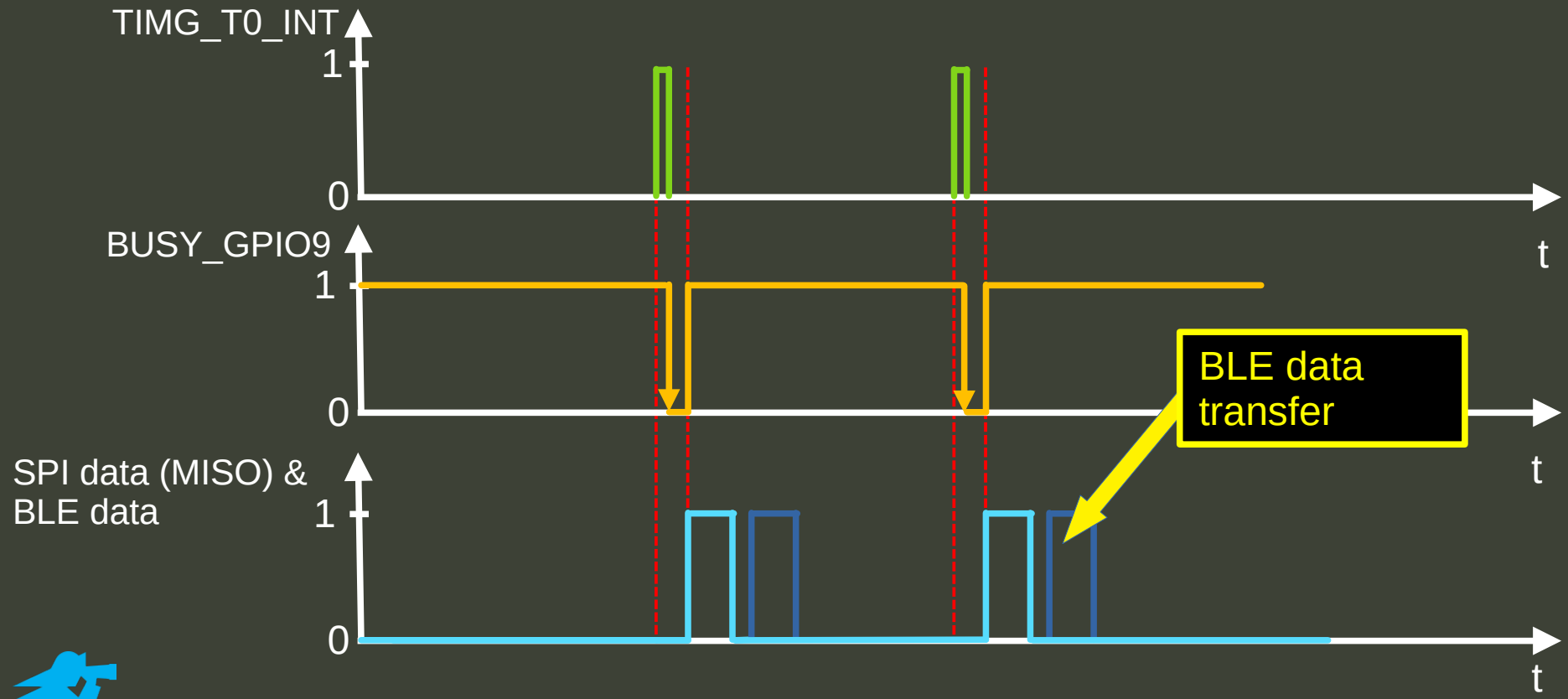
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- ***Loop: Program flow timing diagram (2)***



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- ***Loop: Program flow timing diagram (3)***



Project Seminar - Schedule

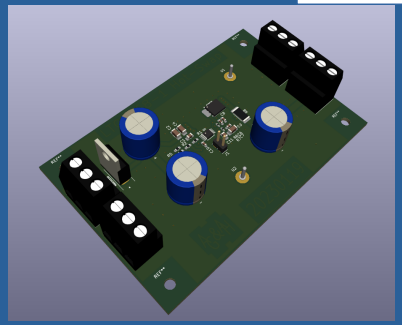


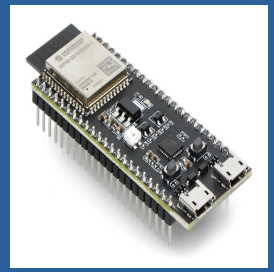
- ***Tasks:***
  - 12.06.2026: independent study – repetition content of lesson 10
  - until 20.06.2026: Exercise 11 → Repeat the exercise 11 - Write down the task of the exercise and the corresponding solution in your report!
  - until 20.06.2026: Exercise 12 → Repeat the exercise 12 - Write down the task of the exercise and the corresponding solution in your report!
  - until 20.06.2026: Submit your complete KiCad-Project-Folder → all symbol libraries, all footprint libraries, all schematics and PCB as Zip-File !!!
  - until 20.06.2026: Submit your complete project report → as PDF-File !!!

Study Regulations (3) – Examination

PCB design

KiCad





solutions

Project documentation

Project report as PDF-file

- ***Schematics, BOM, manufacturing, assembly,***
- ***Solutions to tasks & exercises, etc.***

Put all parts together (PDF & all project releated folders) as Zip-File(s)!

Name, Matr.-No., face-photo of the student on the project cover on cover!

Study Regulations (4) - Exam

- ***Submission date of all project related files and documents:***

Saturday 20th June 2026 until 10 p.m. → deadline !

Project Seminar - SS2026



Thank you !
Have a nice weekend!



